

Foundation (Jalapeno)

- Q1.** What is the factored form of $x^3 + 8$?
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 - 4)$
(D) $(x - 2)(x^2 + 4)$
(E) $(x + 2)(x^2 + 2x + 4)$
- Q2.** A basketball player scores $x^3 - 27$ points in a game. What is the factored form of this expression?
(A) $(x - 3)(x^2 + 3x + 9)$
(B) $(x + 3)(x^2 - 3x + 9)$
(C) $(x - 3)(x^2 - 9)$
(D) $(x + 3)(x^2 + 9)$
(E) $(x + 3)(x^2 - 3x + 9)$
- Q3.** What is the factored form of $64 - x^3$?
(A) $((x - 4)(x^2 + 4x + 16))$
(B) $((x + 4)(x^2 - 4x + 16))$
(C) $((x - 4)(x^2 - 16))$
(D) $((x + 4)(x^2 + 16))$
(E) $((x - 4)(x^2 + 4x + 16))$
- Q4.** The distance between two cyclists is given by $x^3 + 125$. What is the factored form of this expression?
(A) $(x + 5)(x^2 - 5x + 25)$
(B) $(x - 5)(x^2 + 5x + 25)$
(C) $(x + 5)(x^2 + 25)$
(D) $(x - 5)(x^2 - 25)$
(E) $(x - 5)(x^2 + 5x + 25)$
- Q5.** A runner's score is $x^3 - 8$. What is the factored form of this expression?
(A) $(x - 2)(x^2 + 2x + 4)$
(B) $(x + 2)(x^2 - 2x + 4)$
(C) $(x - 2)(x^2 - 4)$
(D) $(x + 2)(x^2 + 4)$
(E) $(x + 2)(x^2 - 2x + 4)$
- Q6.** What is the factored form of $27x^3 + 64$?
(A) $((3x + 4)(9x^2 - 12x + 16))$
(B) $((3x - 4)(9x^2 + 12x + 16))$
(C) $((3x + 4)(9x^2 + 16))$
(D) $((3x - 4)(9x^2 - 16))$
(E) $((3x + 4)(9x^2 - 12x + 16))$
- Q7.** The score of a hockey team is given by $x^3 - 125$. What is the factored form of this expression?
(A) $(x - 5)(x^2 + 5x + 25)$
(B) $(x + 5)(x^2 - 5x + 25)$
(C) $(x - 5)(x^2 - 25)$
(D) $(x + 5)(x^2 + 25)$
(E) $(x + 5)(x^2 + 5x + 25)$
- Q8.** A gymnast's score is $x^3 + 216$. What is the factored form of this expression?
(A) $(x + 6)(x^2 - 6x + 36)$
(B) $(x - 6)(x^2 + 6x + 36)$
(C) $(x + 6)(x^2 + 36)$
(D) $(x - 6)(x^2 - 36)$
(E) $(x - 6)(x^2 + 6x + 36)$

Open Ended Questions (FRQ)

- Q9.** The time it takes for a swimmer to complete a lap is given by $x^3 - 64$. Factor this expression and simplify. Show all steps.
- Q10.** A football team's points are given by $x^3 + 1000$. Factor this expression and simplify. Show all steps.

Chef's Tips

- Chef's Tips
- Always review the factoring rules before the test.
- Provide extra support for students struggling with cube patterns.